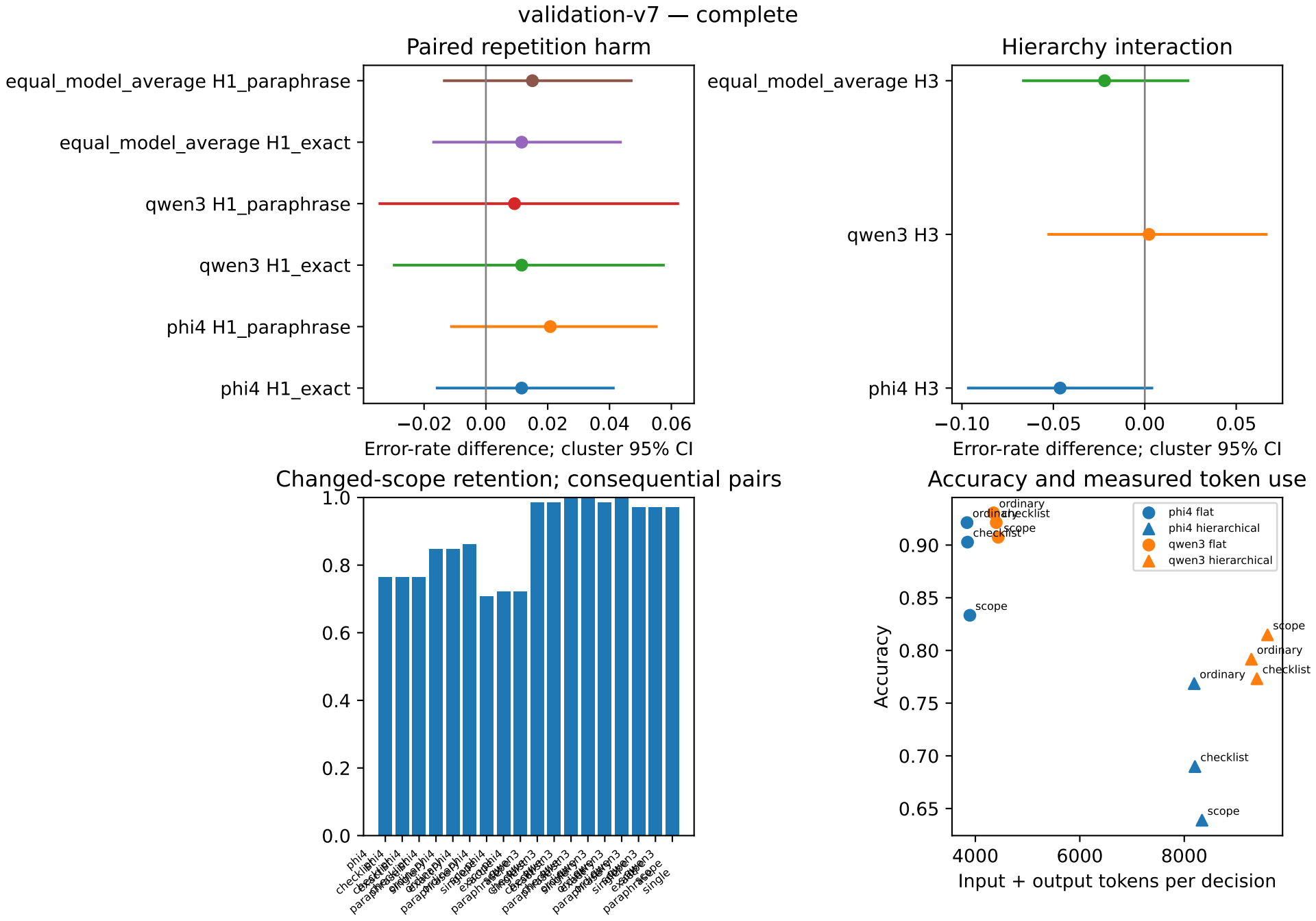
validation-v7 — complete
Paired repetition harm
equal_model_average H1_paraphrase
equal_model_average H1_exact
qwen3 H1_paraphrase
qwen3 H1_exact
phi4 H1_paraphrase
phi4 H1_exact
−0.02
0.00
0.02
0.04
0.06
Error-rate difference; cluster 95% CI
Hierarchy interaction
equal_model_average H3
qwen3 H3
phi4 H3
−0.10
−0.05
0.00
0.05
Error-rate difference; cluster 95% CI
Changed-scope retention; consequential pairs
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy and measured token use
phi4 flat
phi4 hierarchical
qwen3 flat
qwen3 hierarchical
ordinary
checklist
scope
0.90
0.85
0.80
0.75
0.70
0.65
Accuracy
4000
6000
8000
Input + output tokens per decision